

Bounds on the Dynamic Causal Effects of Monetary Policy

An Analysis Using the Federal Reserve's Textual Repository

Felipe Pombal Fritsch

Advisor: Prof. José Luis Montiel Olea

A thesis presented for the degree of Bachelor of Arts
Department of Economics, Columbia University

Submitted May 2020 · Revised edition, September 2026

Abstract

Semiparametric estimates of the effects of monetary policy treat each FOMC meeting as a unit that selects into a policy choice, and require every choice to be possible at every point of the covariate space (overlap). Using the policy alternatives that Federal Reserve staff lay out in the Bluebook before each meeting, I show that overlap fails in a specific, observable way: over 133 meetings from 1989 to 2006 the Committee never took an action that was absent from its menu. I build a propensity-score model in which the menu imposes these structural zeros, derive the sharp identified set for the average dynamic effect of a rate increase or decrease when some menus exclude the action or the status quo, and propose a one-parameter relaxation that trades the worst case for an assumption on effect heterogeneity across menus. The menu constraint alone improves the fit of the policy model as much as adding market expectations. On the empirical side, worst-case bounds are uninformative, and the effects that are point-identified on the meetings with genuine overlap have the wrong sign for industrial production and unemployment even after conditioning on the Greenbook forecasts, although the forecasts do remove the price puzzle. Handling overlap does not rescue selection-on-observables in this setting: the binding problem is unconfoundedness.

Note on this edition. This revision replaces the version submitted in May 2020. The conceptual idea is unchanged; the estimation was revised and the code rebuilt from the raw data. The main changes are the propensity score, the calibration of the outcome bounds, the treatment of the point estimates, and the addition of inference. Several conclusions change as a result. Appendix C compares the two versions. All numbers here are produced by the pipeline in the accompanying repository.

Contents

1	Introduction	3
2	Related Literature	4
2.1	Dynamic causal effects and potential outcomes	4
2.2	Identifying monetary policy shocks and the Federal Reserve's information	4
2.3	The Federal Reserve's text as data	5
2.4	Overlap, weighting and partial identification	5
3	Data: Policy Menus from the Bluebooks	6
3.1	How the FOMC prepares a decision	6
3.2	Extracting the menus	6
3.3	What the menus show	6
4	Framework: Policy Evaluation When Overlap Fails by Design	8
4.1	Treatment, menus and potential outcomes	8
4.2	Assumptions	8
4.3	What the data identify	8
4.4	A sensitivity parameter for the missing menus	9
5	Estimation and Inference	10
5.1	A menu-constrained propensity score	10
5.2	Estimating the bounds	11
5.3	Inference	12
6	Results	12
7	Discussion: The Results in Light of Recent Work	16
8	Conclusion	17
A	Proofs	19
B	Robustness: macro information set	20
C	Comparison with the 2020 version	23

1 Introduction

Measuring the effect of monetary policy on the economy is one of the central problems of empirical macroeconomics. The difficulty is familiar. Policy responds to the state of the economy, which in turn responds to policy, and the econometrician sees a few hundred monthly observations of this feedback loop. The dominant response has been to impose structure, through structural vector autoregressions (Sims, 1980) or local projections (Jordà, 2005), and with it linearity in the effect of policy.

A second approach, developed by Angrist and Kuersteiner (2011) and Angrist et al. (2018), borrows from microeconomic policy evaluation. Each FOMC meeting is a unit that selects into a policy choice (ease, hold, or tighten), and the effect of that choice is recovered by inverse probability weighting (IPW) on a propensity score for the Committee’s decision. The approach is semiparametric and allows effects to be nonlinear and asymmetric. Like all selection-on-observables designs, however, it rests on two assumptions. *Unconfoundedness* requires that, given the observed covariates, the decision be unrelated to the future path the economy would take under each alternative. *Overlap* requires every decision to have positive probability at every value of the covariates.

This thesis starts from the second assumption. Some policy actions are simply not on the table in some states of the economy. The Federal Reserve’s own documents record which: before every meeting, Board staff circulate the Bluebook, which lists the policy alternatives the Committee will consider. I use these *policy menus*, extracted from the Bluebooks by text analysis, to make three contributions.

1. **Structural non-overlap is observable.** In 133 scheduled meetings between August 1989 and January 2006 the FOMC never chose an action that was not on its Bluebook menu, and two thirds of the menus exclude either easing or tightening. The failure of overlap is not a feature of a fitted model. It is recorded in the Committee’s briefing documents.
2. **Menu-constrained propensity score and sharp bounds.** I estimate the decision model as an ordered probit renormalised over the menu, so that the propensity score is exactly zero for an action that is not on the menu. I then derive the sharp identified set for the average effect of a rate increase or decrease, exploiting the fact that in a menu that contains the status quo but not the action (or the reverse) half of the comparison is still identified. A one-parameter relaxation bounds how far effects in the non-overlap menus may differ from the effect in the overlap region, and yields a breakdown value for each estimate’s sign.
3. **Evidence.** Imposing the menus raises the log likelihood of the decision model by almost 19 points without any additional parameters. However, the effects that are point-identified on the overlap region imply that rate increases are followed by higher industrial production and lower unemployment, even when the propensity score conditions on the Greenbook forecasts that Romer and Romer (2004) use to purge policy of its predictable component. Worst-case bounds that make no assumption on the missing menus are wide and contain zero.

The negative result is informative. It shows that correcting the overlap problem, which is real and documented, does not deliver credible effects of monetary policy under selection on observables in this sample: what remains is a failure of unconfoundedness, most plausibly because the Committee acts on information about the future economy that neither macro aggregates, market expectations nor staff forecasts fully capture. The bounds framework makes this transparent, because it separates what the data identify from what must be assumed.

Outline. Section 2 reviews the related literature, including work published since the first version of this thesis. Section 3 describes the FOMC’s decision process and the construction of the policy menus. Section 4 sets out the framework and derives the bounds. Section 5 covers estimation and inference, and Section 6 reports results. Section 7 discusses them in light of recent work, and Section 8 concludes.

2 Related Literature

This thesis sits where four literatures meet: the potential-outcomes approach to dynamic causal effects in macroeconomics, the identification of monetary policy shocks, the use of the Federal Reserve’s text as data, and the econometrics of weighting estimators when overlap is limited or fails. Each has moved substantially since the first version of this thesis was written in 2020.

2.1 Dynamic causal effects and potential outcomes

Angrist and Kuersteiner (2011) and Angrist et al. (2018) recast the effect of monetary policy as a policy-evaluation problem: each FOMC decision is a treatment, a propensity score for the Committee’s choice is estimated by ordered probit, and impulse responses are inverse-probability-weighted averages of future outcomes. Their approach is robust to nonlinearity and asymmetry, and it requires selection on observables and overlap. Rambachan and Shephard (2025) give the general framework: a “direct potential outcome system” in which impulse responses and local projections have a nonparametric causal interpretation under conditions they make precise. This thesis adopts their notation for potential outcomes indexed by the policy action and the horizon.

A closely related line asks what the standard linear tools estimate when the world is not linear. Plagborg-Møller and Wolf (2021) show that local projections and VARs estimate the same population impulse responses, so the choice between them is about finite-sample bias and variance, not about the estimand. Kolesár and Plagborg-Møller (2025) show that VARs and linear local projections on an observed shock or proxy identify *weighted averages* of causal effects however nonlinear the data-generating process, whereas methods that rely on heteroskedasticity or non-Gaussianity for identification can fail. The IPW estimator used here targets an unweighted average effect of a discrete policy action instead. The price is that it needs a correctly specified propensity score and, as this thesis emphasises, overlap. Montiel Olea and Plagborg-Møller (2021) show that lag-augmented local projections give valid inference even with persistent data and at long horizons. Their result is a useful benchmark for the inference problem this thesis faces, in which outcomes at nearby meetings overlap in time.

2.2 Identifying monetary policy shocks and the Federal Reserve’s information

The central difficulty in estimating the effect of monetary policy is that the Federal Reserve reacts to what it expects the economy to do. Romer and Romer (2004) addressed this by purging the intended funds-rate change of its response to the Greenbook forecasts. Romer and Romer (2023) revisit thirty-five years of narrative evidence and argue that carefully identified shocks have large effects on output, unemployment and inflation in the expected directions. High-frequency identification uses the change in market rates in a narrow window around FOMC announcements instead. Nakamura and Steinsson (2018) argue that such surprises also reveal the Fed’s private information about the economy (an “information effect”), and Jarociński and Karadi (2020) and Miranda-Agrippino and

Ricco (2021) separate pure policy shocks from information shocks using the joint response of stock prices or private forecasts. Bauer and Swanson (2023a,b) offer an alternative account: much of the apparent information effect reflects the Fed responding to economic news that markets had not yet priced, which makes raw surprises predictable and calls for orthogonalising them with respect to prior news.

Most closely related to this thesis, Aruoba and Drechsel (forthcoming) use natural language processing on the staff documents prepared before each FOMC meeting to measure the Committee’s information set. They find that the text contains information about the economy that the Greenbook’s numerical forecasts do not, and that shocks identified as residuals from a model that uses it produce responses consistent with standard theory. Their conclusion speaks directly to the empirical results of Section 6: conditioning on numerical forecasts alone may leave enough of the Fed’s information out of the propensity score to bias selection-on-observables estimates.

2.3 The Federal Reserve’s text as data

Hansen et al. (2018) brought computational linguistics to the FOMC transcripts, and the use of the Fed’s internal documents has since grown quickly. Ke et al. (2022) develop robust topic-model estimators and apply them to the transcripts of the Greenspan era, the period studied here. Two recent papers use the Bluebook alternatives themselves. Doh et al. (2026) fine-tune a language model on the alternative policy statements drafted by Board staff to measure where the adopted statement sits on the dovish–hawkish spectrum, and Laarits et al. (2025) use the alternatives circulated before each meeting to study how the Committee aggregates members’ differing economic models. This thesis uses the same documents for a different purpose: not to measure the tone of communication, but to recover the set of actions the Committee considered, which determines where the propensity score is zero.

2.4 Overlap, weighting and partial identification

When propensity scores approach zero, IPW estimators become unstable and standard inference can fail. Khan and Tamer (2010) show that identification is then irregular and convergence slower than $\sqrt{n}$; Crump et al. (2009) propose trimming to a subpopulation with good overlap; Ma and Wang (2020) and Heiler and Kazak (2021) develop inference that remains valid with small or many extreme scores; and D’Amour et al. (2021) show that strict overlap becomes restrictive when there are many covariates. All of these treat limited overlap as a feature of the estimated score. The distinguishing feature of the present setting is that overlap fails *exactly*, and the region where it fails is recorded in the data.

For that case, partial identification is the natural tool. Worst-case bounds under a support restriction go back to Manski (1990, 2003), and Imbens and Manski (2004) give confidence intervals for the partially identified parameter. Susmann et al. (2025) develop bounds on the average treatment effect in the cross-sectional setting when outcome overlap fails for a subpopulation, with width equal to the outcome range times the probability of that subpopulation, and propose efficient estimators and uniformly valid inference. Proposition 1 is the analogue for a multi-valued treatment in a time series, where the non-overlap region is observed rather than estimated and some menus contain only one of the two actions being compared. Finally, a large literature relaxes unconfoundedness itself; Masten and Poirier (2018) bound treatment effects when selection on unobservables is limited by a sensitivity parameter, an approach that complements the λ -relaxation of Section 4, which relaxes effect homogeneity across menus instead.

3 Data: Policy Menus from the Bluebooks

3.1 How the FOMC prepares a decision

The Federal Open Market Committee (FOMC) meets eight times a year at scheduled meetings and occasionally by conference call. Each scheduled meeting is preceded by a structured set of staff documents ([Dankner and Luecke, 2005](#)). About two weeks before the meeting the *Beige Book* summarises anecdotal evidence from the twelve Reserve Bank districts. About a week before, the *Greenbook* gives the Board staff’s forecast for the domestic and international economy. On the Friday before the meeting the *Bluebook* sets out the policy alternatives for the Committee’s consideration, usually labelled A, B and C, each with a proposed federal funds rate target and a rationale. The meeting agenda then includes a discussion of these alternatives before the Committee votes. The format of these documents has changed over time: documents have been introduced, merged and retired since the corpus begins in 1936. Over the sample used here the sequence above is stable.

Two of these documents enter the analysis. The Bluebook defines the *menu* of actions the Committee considered, which is the object of this thesis, and the Greenbook provides the staff forecasts used as covariates in Section 5.

3.2 Extracting the menus

The Bluebooks are published with a five-year lag on the Board’s website as HTML and PDF files. Building on the FOMC text-analysis project of Montiel Olea, Giesecke, Chitale and Frermann, the documents were converted to raw text and every sentence mentioning a policy alternative was extracted with the regular expression

$$([\^.]*) (alternatives?\s+[a-e]) (\. | [\^a-z] \. | [\^a-z] [\^\.] + \.)$$

which captures a sentence containing “alternative” followed by a letter. The extracted sentences were then read and classified by hand, because the same target is sometimes described in different ways under different rationales. For each alternative I record the proposed change in the federal funds rate target, and classify it as easing (*E*), unchanged (*U*) or tightening (*T*).

The Committee’s actual decision is taken from the target changes in the replication files of [Angrist et al. \(2018\)](#). The sample runs from August 1989 to January 2006, the Greenspan chairmanship after the Committee had begun to target the federal funds rate explicitly ([Romer and Romer, 2004](#)). It contains 133 scheduled meetings with a Bluebook menu. Target changes of 25, 50 and 75 basis points are pooled into the three categories because 50 and 75 basis point moves are too rare to estimate separately.

3.3 What the menus show

Table 1 cross-tabulates the menu offered with the decision taken. Two facts stand out. First, the decision is *always* on the menu: no meeting in the sample took an action that the Bluebook did not propose. The zeros in Table 1 are therefore structural, and they are observed rather than inferred. Second, most menus are restricted: 87 of 133 meetings (65%) were offered a menu without easing or without tightening, and only 46 had all three options. The full menu is also where the Committee was least likely to move. Only one of the 27 tightenings in the sample came from a full menu.

Table 1: Bluebook menus and FOMC decisions, August 1989 – January 2006

Bluebook menu X_t	Ease	Unchanged	Tighten	Meetings
$\{E\}$	3	0	0	3
$\{U\}$	0	1	0	1
$\{T\}$	0	0	7	7
$\{U, E\}$	17	16	0	33
$\{U, T\}$	0	24	19	43
$\{U, T, E\}$	9	36	1	46
Total	29	77	27	133

Notes: Scheduled FOMC meetings. Rows give the set of alternatives proposed in the Bluebook (E ease, U unchanged, T tighten); columns give the decision. Every decision lies in the menu.

Figure 1 shows where in the macroeconomic state space these restrictions bind. Panel (a) plots lagged inflation and industrial-production growth at each meeting, marking the meetings at which the Committee tightened or eased. Panel (b) marks instead the meetings whose menu *excluded* easing or tightening. Menus without easing are concentrated where output growth is above its sample average (88% of them), and menus without tightening where it is below (78%). The same pattern appears in the decisions themselves, but the menu shows it without relying on a fitted model: in about half of the meetings with below-average output growth (29 of 56) tightening was not even proposed, and the Committee tightened at only three of them. This is exactly the kind of covariate region in which a propensity score must be zero, and it is why a standard IPW estimator, which divides by the propensity score, is not available there.

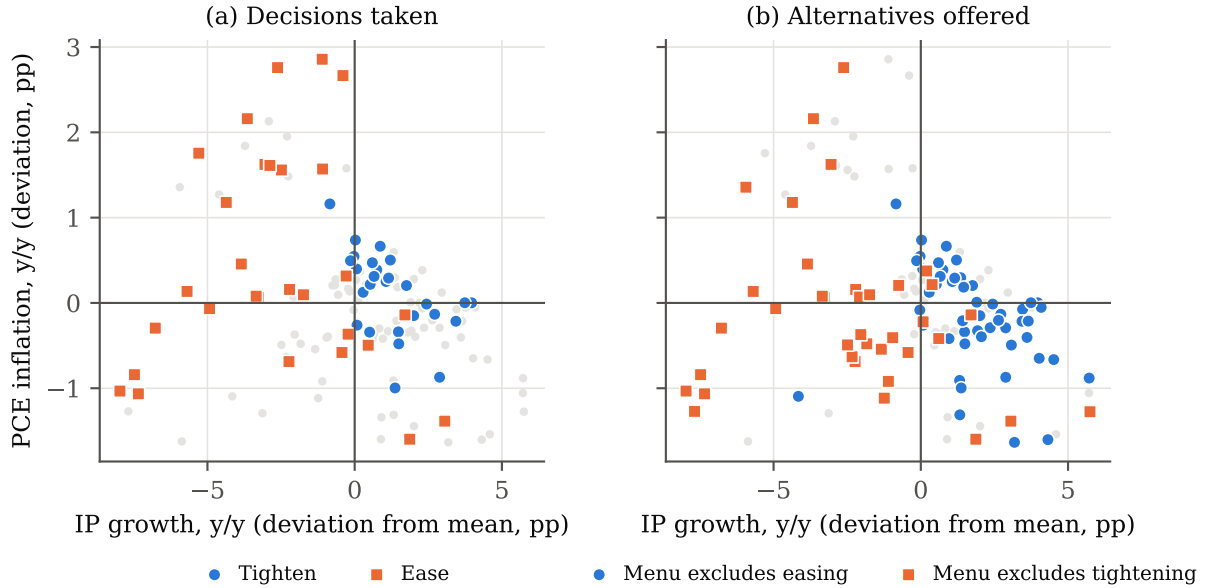


Figure 1: Policy decisions and policy menus in the inflation–output space. Each point is a scheduled FOMC meeting; axes show year-on-year PCE inflation and industrial-production growth in the previous month, as deviations from their 1989–2006 means. Grey points are meetings not in the highlighted group.

4 Framework: Policy Evaluation When Overlap Fails by Design

4.1 Treatment, menus and potential outcomes

Index scheduled FOMC meetings by t . The policy decision is $D_t \in \mathcal{D} = \{-1, 0, 1\}$ (ease, leave unchanged, tighten the federal funds rate target). The Bluebook menu is $X_t \in \mathcal{X}$, a non-empty subset of $\mathcal{D}$; in the data $\mathcal{X} = \{\{-1\}, \{0\}, \{1\}, \{-1, 0\}, \{0, 1\}, \{-1, 0, 1\}\}$. Other information available to the Committee (lagged macroeconomic data, market expectations, staff forecasts) is collected in $Z_t \in \mathbb{R}^k$, and $W_t = (X_t, Z_t)$.

For an outcome series y (log industrial production, the unemployment rate, the log PCE price index), let $Y_{t,h} = y_{t+h} - y_t$ be its cumulative change over the h months following the meeting. Each decision d induces a potential outcome $Y_{t,h}(d)$, and $Y_{t,h} = \sum_{d \in \mathcal{D}} \mathbf{1}\{D_t = d\} Y_{t,h}(d)$. The parameter of interest is the average dynamic effect of an increase ($d = 1$) or a decrease ($d = -1$) relative to leaving the target unchanged,

$$\theta_d(h) = \mathbb{E}[Y_{t,h}(d) - Y_{t,h}(0)], \quad d \in \{-1, 1\}, \quad h = 1, \dots, 24. \quad (1)$$

Expectations are taken over the stationary distribution of meetings. I drop the t and h subscripts when they are not needed. The propensity score is $p_d(w) = \mathbb{P}(D = d \mid W = w)$.

4.2 Assumptions

Assumption 1 (Selection on observables). $Y_h(d) \perp\!\!\!\perp D \mid W$ for all $d \in \mathcal{D}$ and all h .

Assumption 2 (Menu overlap). For all $w = (x, z)$ in the support of W : $p_d(x, z) > 0$ if and only if $d \in x$.

Assumption 3 (Bounded outcomes). $Y_h(d) \in [K_{1h}, K_{2h}]$ almost surely, for all d and h , with known $K_{1h} < K_{2h}$.

Assumption 1 is the identifying assumption of Angrist et al. (2018). It says that once we condition on what the Committee knew, including what it was offered, the remaining variation in its decision is as good as random with respect to the economy’s future path. Assumption 2 replaces the usual overlap condition, $p_d(w) > 0$ for all d and w . It allows the propensity score to be zero, but only in the way the Bluebook dictates: the Committee does not take an action that is not on the menu (which holds without exception in the data, Table 1), and any action on the menu has some chance of being taken. The “only if” part is testable and holds. The “if” part is an assumption. Assumption 3 is a support restriction of the kind used by Manski (1990). It bites only on the cells in which the data are silent.

4.3 What the data identify

Write $\pi_x = \mathbb{P}(X = x)$ and $\mu_d(x) = \mathbb{E}[Y(d) \mid X = x]$. Conditioning on the menu,

$$\theta_d = \sum_{x \in \mathcal{X}} \pi_x (\mu_d(x) - \mu_0(x)). \quad (2)$$

Under Assumptions 1–2, for any $a \in x$ the conditional mean $\mu_a(x)$ is identified by inverse probability weighting within the menu:

$$\mu_a(x) = \mathbb{E} \left[\frac{\mathbf{1}\{D = a\} Y}{p_a(X, Z)} \mid X = x \right], \quad a \in x, \quad (3)$$

because $p_a(x, Z) > 0$ for every Z when $a \in x$. When $a \notin x$, the data never show $Y(a)$ in menu x and $\mu_a(x)$ is not identified. For a given d , partition the menus into four groups:

$$\begin{aligned}\mathcal{A}_d &= \{x : d \in x, 0 \in x\}, & \mathcal{B}_d &= \{x : 0 \in x, d \notin x\}, \\ \mathcal{C}_d &= \{x : d \in x, 0 \notin x\}, & \mathcal{N}_d &= \{x : d \notin x, 0 \notin x\}.\end{aligned}$$

For $d = 1$ these are $\mathcal{A}_1 = \{\{0, 1\}, \{-1, 0, 1\}\}$, $\mathcal{B}_1 = \{\{0\}, \{-1, 0\}\}$, $\mathcal{C}_1 = \{\{1\}\}$ and $\mathcal{N}_1 = \{\{-1\}\}$. The overlap region $\mathcal{A}_d$ is where the effect is point-identified. In $\mathcal{B}_d$ the status-quo mean is identified but the treated mean is not; in $\mathcal{C}_d$ the reverse; in $\mathcal{N}_d$ neither.

Proposition 1 (Sharp identified set). *Under Assumptions 1–3, the identified set for $\theta_d(h)$ is the interval $[L_d, U_d]$ with*

$$\begin{aligned}L_d &= \sum_{x \in \mathcal{A}_d} \pi_x(\mu_d(x) - \mu_0(x)) + \sum_{x \in \mathcal{B}_d} \pi_x(K_1 - \mu_0(x)) \\ &\quad + \sum_{x \in \mathcal{C}_d} \pi_x(\mu_d(x) - K_2) + \sum_{x \in \mathcal{N}_d} \pi_x(K_1 - K_2), \\ U_d &= \sum_{x \in \mathcal{A}_d} \pi_x(\mu_d(x) - \mu_0(x)) + \sum_{x \in \mathcal{B}_d} \pi_x(K_2 - \mu_0(x)) \\ &\quad + \sum_{x \in \mathcal{C}_d} \pi_x(\mu_d(x) - K_1) + \sum_{x \in \mathcal{N}_d} \pi_x(K_2 - K_1).\end{aligned}$$

Its width is $U_d - L_d = (K_2 - K_1)[\pi(\mathcal{B}_d) + \pi(\mathcal{C}_d) + 2\pi(\mathcal{N}_d)]$.

The proof is in Appendix A. The width is always below the width of the no-information bound, $2(K_2 - K_1)$, so the set is informative whenever some menu contains both d and 0 or one of them. It is also narrower than the naive bound that treats every cell outside $\mathcal{A}_d$ as uninformative, whose width is $2(K_2 - K_1)[1 - \pi(\mathcal{A}_d)]$, because the identified half of each one-sided cell is used. The 2020 version of this thesis used the naive width.

Remark 1 (Additional covariates). Conditioning on Z in (3) is needed for Assumption 1, but the bounds only require the menu-level means. If X happens to contain all the information in Z that predicts D , the propensity score does not depend on Z and (3) reduces to a difference in means within menu; nothing in Proposition 1 changes (Appendix A).

4.4 A sensitivity parameter for the missing menus

The worst case in Proposition 1 lets the average effect in menus outside the overlap region be as extreme as the outcome's support permits. A more useful statement bounds how different those effects can be from the identified effect in the overlap region,

$$\theta_d^A = \mathbb{E}[Y(d) - Y(0) \mid X \in \mathcal{A}_d] = \frac{\sum_{x \in \mathcal{A}_d} \pi_x(\mu_d(x) - \mu_0(x))}{\pi(\mathcal{A}_d)}.$$

Assumption 4 (λ -bounded heterogeneity). *For some $\lambda \geq 0$ and all $x \notin \mathcal{A}_d$: $|\mathbb{E}[Y(d) - Y(0) \mid X = x] - \theta_d^A| \leq \lambda$.*

Proposition 2. *Under Assumptions 1–4 the identified set is obtained from Proposition 1 by intersecting, for each $x \notin \mathcal{A}_d$, the cell-level interval with $[\theta_d^A - \lambda, \theta_d^A + \lambda]$. When the support constraint does not bind this is*

$$\Theta_d(\lambda) = \left[\theta_d^A - \lambda(1 - \pi(\mathcal{A}_d)), \theta_d^A + \lambda(1 - \pi(\mathcal{A}_d)) \right],$$

and the sign of θ_d is identified if and only if $\lambda < \lambda_d^* = |\theta_d^A|/(1 - \pi(\mathcal{A}_d))$.

At $\lambda = 0$ effects are homogeneous across menus and $\theta_d = \theta_d^A$; as λ grows the set widens until it reaches the worst case. The *breakdown value* λ_d^* is the smallest degree of heterogeneity across menus that would overturn the sign of the estimated effect, measured in the units of the outcome. It gives a transparent way to judge how much the conclusion depends on extrapolating from the overlap region.

5 Estimation and Inference

5.1 A menu-constrained propensity score

Following Angrist and Kuersteiner (2011) and Angrist et al. (2018), the Committee’s decision is modelled as an ordered probit in a latent policy index $Z'\beta$ with cutoffs $c_1 < c_2$, which gives unconstrained cell probabilities $\tilde{p}_{-1}(z) = \Phi(c_1 - z'\beta)$, $\tilde{p}_0(z) = \Phi(c_2 - z'\beta) - \Phi(c_1 - z'\beta)$ and $\tilde{p}_1(z) = 1 - \Phi(c_2 - z'\beta)$. To impose Assumption 2 exactly, the probabilities are renormalised over the menu:

$$p_d(x, z) = \frac{\tilde{p}_d(z) \mathbf{1}\{d \in x\}}{\sum_{d' \in x} \tilde{p}_{d'}(z)}. \quad (4)$$

This is the probability of choosing d from an ordered probit whose outcome is known to lie in x . It is zero exactly off the menu, it reduces to the ordered probit when $x = \mathcal{D}$, and it is degenerate at one in the single-option menus, which therefore do not enter the likelihood. The parameters (β, c_1, c_2) are estimated by maximum likelihood; standard errors come from the inverse Hessian, and average marginal effects use the delta method.

The 2020 version entered the menus as dummy variables in an unconstrained ordered probit, which keeps the probabilities off the menu small but positive; the renormalisation makes them exactly zero, as the bounds assume.

Covariates. I consider two information sets. The *macro* set follows Angrist et al. (2018): the surprise component of federal funds futures prices, two lags of monthly PCE inflation, two lags of the change in the unemployment rate, and two lags of year-on-year industrial-production growth. (The 2020 version used the contemporaneous level of the IP index; growth rates lagged to the meeting date are used here instead.) The *Greenbook* set, which is the main specification, replaces the macro variables with the staff forecasts the Committee had in hand at the meeting, as in Romer and Romer (2004): current- and next-quarter forecasts of real GDP growth and GDP-deflator inflation, and the current-quarter unemployment forecast, together with the futures surprise. Each meeting is matched to the most recent Greenbook released before it.

Table 2 reports average marginal effects on the probability of a rate increase. Columns (1)–(2) reproduce the progression of the 2020 thesis on the meeting sample: adding the futures surprise raises the log likelihood from -110.0 to -91.4 . Column (3) imposes the menus through (4), with the same covariates and no additional parameter, and the log likelihood rises by a similar amount again, to -72.6 . The text of the Bluebook carries as much information about the decision as market expectations do. Column (4), the main specification, uses the Greenbook information set and fits best (-52.7). Higher current-quarter forecasts of growth and inflation raise the probability of a hike. The next-quarter inflation forecast enters negatively once the current-quarter forecast is held

fixed; the two are correlated (0.63), so their separate coefficients are less sharply determined than their sum. The futures surprise remains the strongest single predictor.

Table 2: Propensity-score models: average marginal effects on $\mathbb{P}(D_t = 1)$

	(1) Macro	(2) + FFF	(3) + Menus	(4) Greenbook
Inflation, lag 1	−0.056 (0.171)	−0.077 (0.150)	−0.013 (0.129)	
Inflation, lag 2	0.167 (0.151)	−0.039 (0.134)	−0.021 (0.114)	
Δ Unemployment, lag 1	−0.332 (0.203)	−0.239 (0.176)	−0.145 (0.142)	
Δ Unemployment, lag 2	−0.505** (0.226)	−0.239 (0.197)	−0.040 (0.159)	
IP growth, lag 1	0.075** (0.038)	0.059* (0.032)	0.041 (0.027)	
IP growth, lag 2	−0.037 (0.037)	−0.027 (0.032)	−0.022 (0.026)	
Fed funds futures surprise		0.738*** (0.119)	0.636*** (0.114)	0.575*** (0.082)
GB real GDP growth, q_0				0.049*** (0.014)
GB real GDP growth, q_1				0.019 (0.016)
GB inflation, q_0				0.057*** (0.022)
GB inflation, q_1				−0.097*** (0.026)
GB unemployment, q_0				0.005 (0.015)
Menu constraint	No	No	Yes	Yes
Log likelihood	−109.95	−91.41	−72.61	−52.65
Meetings	133	133	133	133

Notes: Ordered probit for the FOMC decision at 133 scheduled meetings, August 1989 – January 2006. Columns (3)–(4) renormalise the probabilities over the Bluebook menu, equation (4). Delta-method standard errors in parentheses. *, **, ***: significant at 10%, 5%, 1%. GB = Greenbook forecast; q_0 , q_1 = current and next quarter.

5.2 Estimating the bounds

Given fitted scores $\hat{p}$, the conditional means in (3) are estimated with normalised (Hájek) inverse probability weights within each menu,

$$\hat{\mu}_a(x) = \frac{\sum_{t: X_t=x} \mathbf{1}\{D_t = a\} Y_{t,h} / \hat{p}_a(W_t)}{\sum_{t: X_t=x} \mathbf{1}\{D_t = a\} / \hat{p}_a(W_t)},$$

and π_x by the sample share of meetings with menu x . Plugging these into Proposition 1 gives $\hat{L}_d$, $\hat{U}_d$ and $\hat{\theta}_d^A$. Normalised weights are bounded by the observed outcomes and are less sensitive to

small scores than the unnormalised estimator. In the main specification the smallest estimated probability of an observed decision is 0.054, so no trimming is needed.

The outcome bounds K_{1h}, K_{2h} are set to the smallest and largest realised h -month change in each outcome over August 1989 – December 2010, a period that includes the 2001 and 2008–09 recessions. The assumption is that the potential outcomes of any single meeting stay within the range the economy actually went through over these two decades. The 2020 version calibrated K to the interquartile range of the estimated effects across horizons; calibrating it to the range of the outcome itself matches Assumption 3 and gives considerably wider bounds.

5.3 Inference

The estimators are smooth functions of sample moments and of the probit estimates, but the observations are serially dependent and the outcomes of nearby meetings overlap. I use a moving-block bootstrap over meetings (Künsch, 1989) with blocks of eight consecutive meetings (about one year) and $B = 499$ draws. Each draw re-estimates the propensity score, the menu shares and the conditional means, so first-stage uncertainty is propagated. The 2020 version left inference for future work because the conditional effects are computed on non-consecutive subsamples; resampling the full sequence of meetings and recomputing everything in each draw addresses that concern.

For θ_d^A I report the basic bootstrap 90% interval. For the partially identified θ_d I report the Imbens and Manski (2004) interval, which covers the true parameter (rather than the whole identified set) with probability 0.9 and remains valid when the identified set is narrow:

$$\left[\hat{L}_d - c_\alpha \hat{\sigma}_L, \hat{U}_d + c_\alpha \hat{\sigma}_U \right], \quad \Phi\left(c_\alpha + \frac{\hat{U}_d - \hat{L}_d}{\max(\hat{\sigma}_L, \hat{\sigma}_U)}\right) - \Phi(-c_\alpha) = 0.9,$$

with $\hat{\sigma}_L, \hat{\sigma}_U$ the bootstrap standard errors. The support bounds K are treated as known.

6 Results

Figure 2 and Table 3 report, for each outcome and each policy, three objects: the worst-case identified set of Proposition 1 (grey), the effect on the overlap region $\hat{\theta}_d^A$ with its 90% bootstrap interval (blue), and the breakdown value λ_d^* of Proposition 2. Responses are cumulative changes from the month of the meeting: percent for industrial production and the PCE price index, percentage points for the unemployment rate. The overlap region covers 67% of meetings for a rate increase (menus $\{U, T\}$ and $\{U, T, E\}$) and 59% for a decrease ($\{U, E\}$ and $\{U, T, E\}$). The main specification uses the Greenbook propensity score.

Worst-case bounds are not informative. With K equal to the range of outcomes the economy actually experienced, every identified set contains zero at every horizon, and so does every Imbens and Manski (2004) interval. Two years after a rate increase, the identified set for industrial production runs from -7.2% to $+4.3\%$. The menus that exclude the action or the status quo account for a third or more of all meetings, and without an assumption about them the data cannot sign the average effect. The 2020 version reported much narrower bounds because it calibrated K differently (Appendix C).

The overlap-region effects have the wrong sign for real activity. Where both the action and the status quo were on the menu, the effects are point-identified and estimated with some

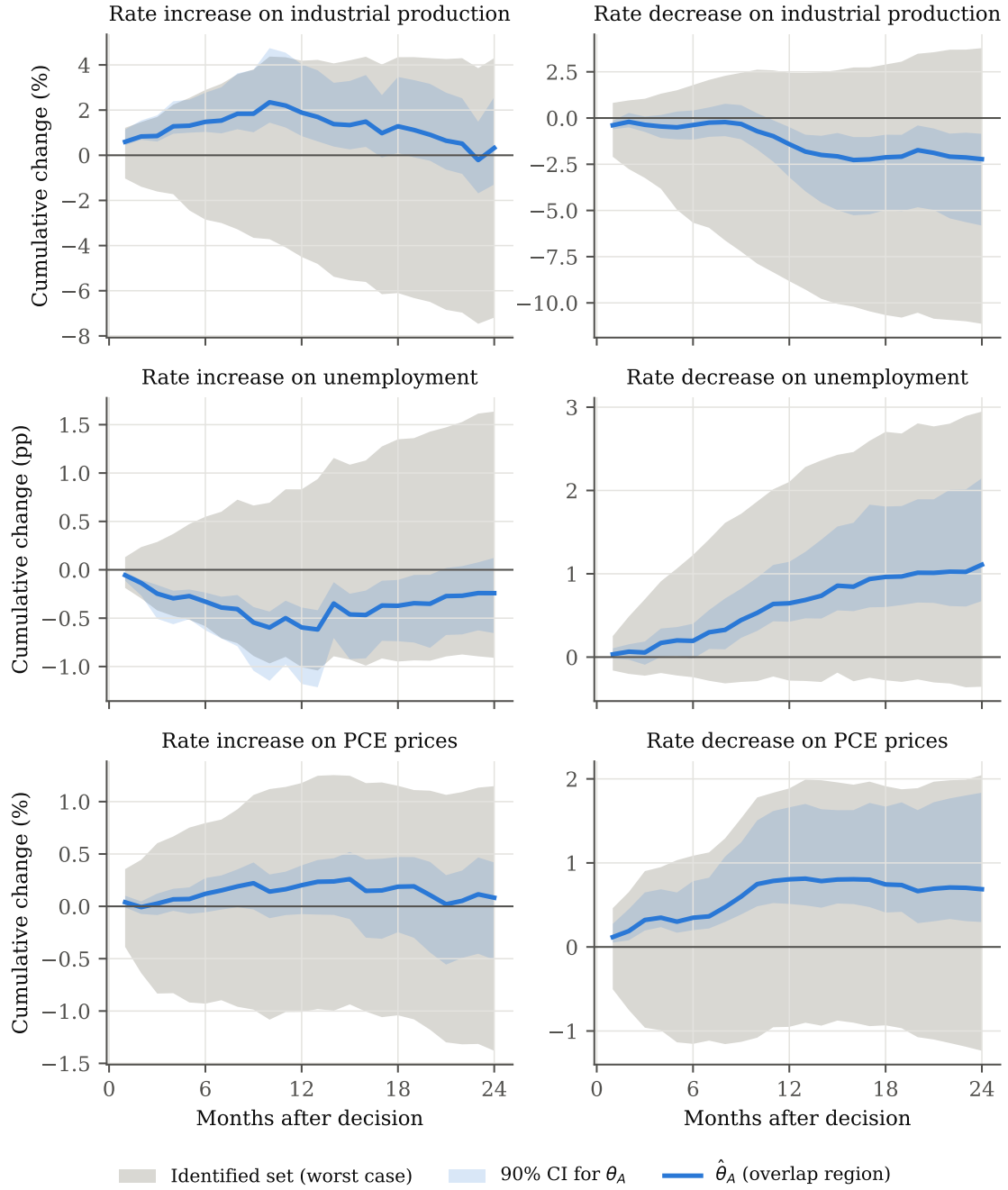


Figure 2: Cumulative effects of a rate increase (left) and decrease (right) relative to no change. Grey: worst-case identified set (Proposition 1). Blue: effect on the overlap region $\hat{\theta}_d^A$ with 90% moving-block bootstrap interval. Greenbook propensity score; 133 meetings, 1989–2006.

precision. After a rate *increase*, industrial production is higher than under no change (by 1.5% at six months and 1.9% at twelve, with 90% intervals excluding zero through 18 months) and unemployment is lower (by 0.6 percentage points at twelve months). After a rate *decrease*, industrial production is lower by 1.4% at twelve months and 2.2% at twenty-four, and unemployment is higher by 0.65 and 1.1 points. These are the opposite of the responses implied by standard theory and found by most of the literature, and they are the same in the macro specification (Appendix B).

Staff forecasts remove the price puzzle. The price responses behave differently. With the macro information set, a rate increase is followed by significantly *higher* prices from five months on, which is the well-known price puzzle (Sims, 1992). Conditioning on the Greenbook forecasts makes the price response to a rate increase small and statistically indistinguishable from zero at every horizon (0.20% at twelve months, interval $[-0.07, 0.39]$). This is the pattern Romer and Romer (2004) report: the puzzle reflects the Committee’s reaction to inflation it foresees, and the staff forecast captures that information. A rate decrease is followed by higher prices under both specifications (0.8–1.0% at twelve months), the direction theory predicts.

Table 3: Effects of rate changes relative to no change, Greenbook information set

Policy	h	$\hat{\theta}_A$	90% CI	Identified set	λ^*
<i>Industrial production (%)</i>					
Rate increase	6	1.48	[1.03, 2.77]	$[-2.8, 2.9]$	4.47
	12	1.89	[0.85, 4.02]	$[-4.5, 4.2]$	5.71
	18	1.28	[0.10, 3.47]	$[-6.1, 4.3]$	3.88
	24	0.32	$[-1.29, 2.58]$	$[-7.2, 4.3]$	0.97
Rate decrease	6	-0.38	$[-1.17, 0.41]$	$[-5.7, 1.8]$	0.93
	12	-1.42	$[-3.20, -0.49]$	$[-8.8, 2.5]$	3.49
	18	-2.12	$[-4.98, -0.90]$	$[-10.7, 2.9]$	5.23
	24	-2.22	$[-5.82, -0.85]$	$[-11.1, 3.8]$	5.48
<i>Unemployment rate (pp)</i>					
Rate increase	6	-0.33	$[-0.62, -0.24]$	$[-0.6, 0.5]$	1.00
	12	-0.60	$[-1.18, -0.39]$	$[-1.0, 0.8]$	1.80
	18	-0.37	$[-0.74, -0.11]$	$[-0.9, 1.3]$	1.12
	24	-0.24	$[-0.65, 0.12]$	$[-0.9, 1.6]$	0.73
Rate decrease	6	0.19	$[-0.02, 0.40]$	$[-0.2, 1.2]$	0.48
	12	0.65	[0.42, 1.14]	$[-0.3, 2.1]$	1.59
	18	0.96	[0.60, 1.81]	$[-0.3, 2.7]$	2.37
	24	1.11	[0.67, 2.15]	$[-0.4, 2.9]$	2.73
<i>PCE price index (%)</i>					
Rate increase	6	0.12	$[-0.06, 0.27]$	$[-0.9, 0.8]$	0.36
	12	0.20	$[-0.07, 0.39]$	$[-1.0, 1.2]$	0.61
	18	0.19	$[-0.25, 0.47]$	$[-1.0, 1.2]$	0.57
	24	0.08	$[-0.51, 0.42]$	$[-1.4, 1.1]$	0.25
Rate decrease	6	0.35	[0.20, 0.78]	$[-1.2, 1.1]$	0.86
	12	0.80	[0.51, 1.66]	$[-1.0, 1.9]$	1.98
	18	0.74	[0.41, 1.67]	$[-0.9, 1.9]$	1.83
	24	0.69	[0.30, 1.84]	$[-1.2, 2.0]$	1.69

Notes: Cumulative change h months after the meeting. $\hat{\theta}_A$: effect on meetings whose menu contains both the action and no change, with basic moving-block bootstrap 90% interval (499 draws, blocks of 8 meetings). Identified set: worst-case bounds on the average effect over all meetings (Proposition 1). λ^* : smallest heterogeneity between the non-overlap menus and the overlap region, in the outcome’s units, that would make the sign of the average effect unidentified (Proposition 2).

How much extrapolation is needed? If effects were the same in every menu ($\lambda = 0$), the overlap-region estimates would be the average effects. The breakdown values in Table 3 say how far that homogeneity can be relaxed before the sign of the average effect is lost. For a rate increase, a third of the meetings are outside the overlap region, so $\lambda^* = 3|\hat{\theta}^A|$: the twelve-month IP response keeps its sign unless the average effect in the menus without the status quo or without tightening differs from the overlap-region effect by more than 5.7 percentage points. For a rate decrease two fifths of meetings lie outside the region and $\lambda^* \approx 2.4|\hat{\theta}^A|$. Figure 3 traces the full set as a function of λ . Reconciling the average effects with the textbook signs would require the effect in the non-overlap menus to be not merely different from, but several times larger than and opposite in sign to, the effect in the overlap region. That cannot be ruled out, but it would be an odd defence: the counter-theoretical signs come from the meetings where the data are most informative, not from extrapolation.

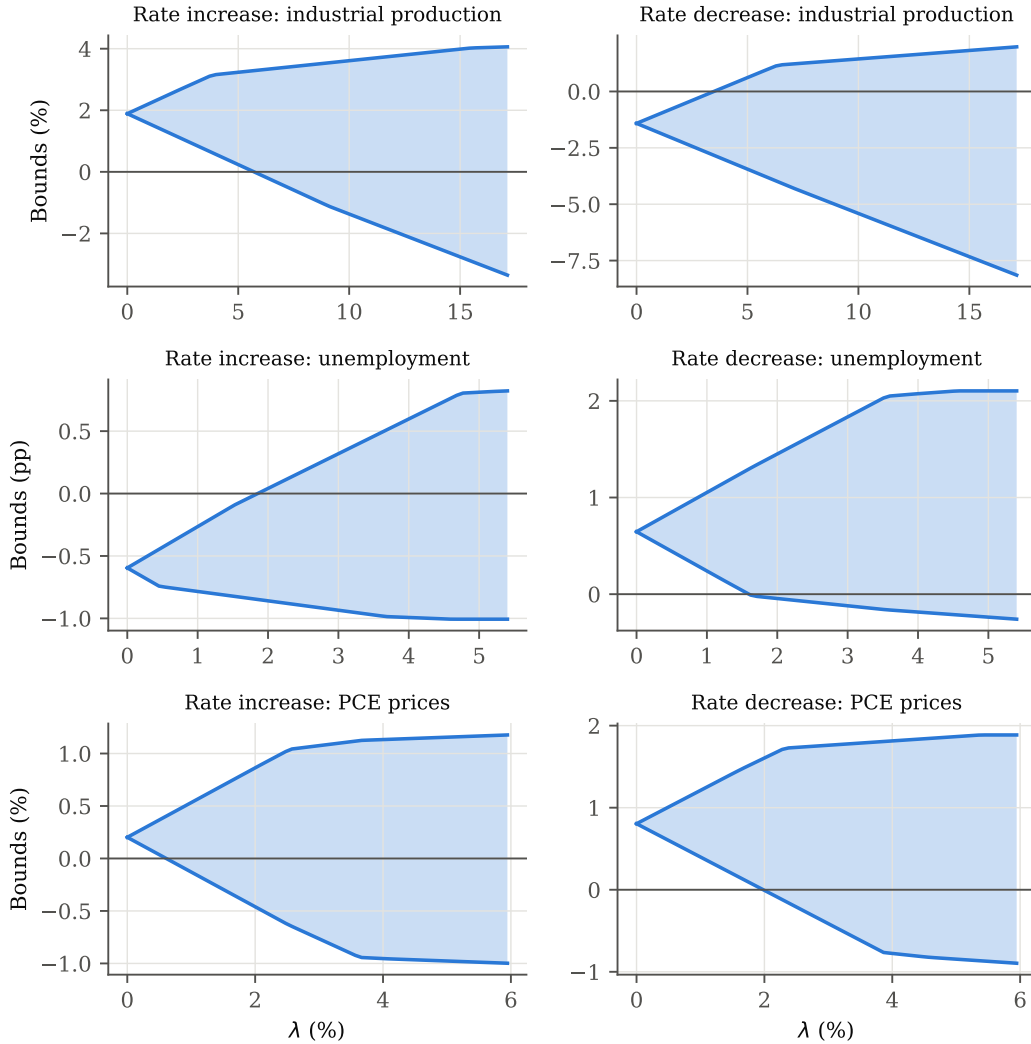


Figure 3: Identified set for the twelve-month effect as a function of λ , the maximal difference between the average effect in a non-overlap menu and in the overlap region (Assumption 4). At $\lambda = 0$ the set is the point $\hat{\theta}_d^A$; for large λ it reaches the worst case. Greenbook propensity score.

Interpretation. The overlap-region estimates are credible only under Assumption 1. Their signs indicate that it fails. The Committee tightens when it expects the economy to strengthen, and the expected strength shows up in subsequent output whether or not the Committee acts. Neither lagged data, market expectations nor the Greenbook forecasts absorb this completely. Two features of the sample make the problem hard to escape. First, the identifying variation is thin: all but one of the 27 tightenings come from $\{U, T\}$ menus, so the effect of a rate increase is essentially a comparison of hikes and holds within those 43 meetings. Second, weighting within a menu only reweights meetings the Committee itself judged similar enough to offer the same options, so the remaining differences between hikes and holds are exactly the judgemental information that is hardest to measure. Asymmetries between increases and decreases, which the 2020 version emphasised following Hamilton and Jordà (2002) and Angrist and Kuersteiner (2011), cannot be separated from this selection problem with the present data.

7 Discussion: The Results in Light of Recent Work

Read against the literature of Section 2, the results support three conclusions and point to three extensions.

The counter-theoretical signs are the expected symptom of an incomplete information set. The overlap-region estimates imply that rate increases are followed by stronger activity. That is what one would see if the Committee tightened when it had reason to expect strength that the econometrician’s covariates do not capture. This is the mechanism behind the information effect of Nakamura and Steinsson (2018) and, in the account of Bauer and Swanson (2023a), behind the predictability of market surprises by economic news. The finding that Greenbook forecasts remove the price puzzle but not the real-activity puzzle fits the evidence in Aruoba and Drechsel (forthcoming) that the staff’s written analysis contains information not summarised by its numerical forecasts. It also fits the argument of Romer and Romer (2023) that isolating policy changes unrelated to the outlook requires reading the record, not only conditioning on forecasts. The menus do not solve this problem, and they are not designed to. What they do is separate it cleanly from the overlap problem, so that the remaining bias can be attributed to unconfoundedness.

The menu restriction is a genuine source of identifying information. Imposing the menus raises the likelihood of the decision model by as much as market expectations do, and it is exact rather than approximate. Recent work that uses the same documents for other purposes (Doh et al., 2026; Laarits et al., 2025) confirms that the Bluebook alternatives are a well-defined and informative record of the Committee’s deliberations. In the terms of the overlap literature, the menus turn a problem of extreme estimated scores (Khan and Tamer, 2010; Heiler and Kazak, 2021) into one of structural zeros in a known region. That region can be handled by partial identification (Susmann et al., 2025) instead of trimming, which would change the estimand.

Worst-case bounds alone are too wide for macroeconomic questions. With one third to two fifths of meetings outside the overlap region and outcome ranges that include the 2008–09 recession, the identified sets contain zero everywhere. The same arithmetic makes the cross-sectional bounds of Susmann et al. (2025) informative only when the non-overlap subpopulation is small. In monetary policy it is not, so informative conclusions require an additional assumption. The breakdown values λ^* make that assumption explicit and give it units.

Extensions. First, the menu-constrained propensity score can be combined with a richer information set. Text-based measures of the staff’s outlook in the spirit of [Aruoba and Drechsel \(forthcoming\)](#) are the obvious candidate, and they can be built from the same Bluebook and Greenbook corpus used here. Second, the λ -relaxation of effect homogeneity can be paired with a relaxation of unconfoundedness along the lines of [Masten and Poirier \(2018\)](#). That would report how much selection on unobservables is needed to overturn each sign, which is the question the results raise. Third, the menus can be used with external shocks. [Kolesár and Plagborg-Møller \(2025\)](#) show that linear projections on an observed shock identify a weighted average of causal effects even when the economy is nonlinear. Estimating such projections within menus, with high-frequency surprises purged of prior news as in [Bauer and Swanson \(2023b\)](#), would combine the structural restriction documented here with a source of variation that does not rely on selection on observables.

8 Conclusion

This thesis set out to relax one of the two assumptions behind semiparametric estimates of the effects of monetary policy. Its central observation holds up: the Federal Reserve’s own briefing documents show that some actions are not on the table at some meetings, and the Committee never chose an action its staff had not proposed. Treating the Bluebook menu as a covariate that dictates where the propensity score is zero gives a decision model that fits markedly better at no cost in parameters, and a sharp characterisation of what the data can and cannot say about the average effect of policy.

What the data can say is limited. Without restrictions on the menus that exclude the action or the status quo, the average effect is not signed. With effects assumed similar across menus, the estimates are precise but have the wrong sign for output and unemployment, even with the Greenbook forecasts in the propensity score. The forecasts do remove the price puzzle for rate increases. The contribution of the menu is therefore to isolate the problem. Overlap can be handled; unconfoundedness is what fails.

Section 7 sets out the natural next steps: a richer, text-based information set inside the menu-constrained propensity score, a sensitivity analysis for unconfoundedness alongside the one for effect homogeneity, and the combination of the menu restriction with high-frequency policy surprises.

References

- Joshua D. Angrist and Guido M. Kuersteiner. Causal effects of monetary shocks: Semiparametric conditional independence tests with a multinomial propensity score. *Review of Economics and Statistics*, 93(3):725–747, 2011.
- Joshua D. Angrist, Òscar Jordà, and Guido M. Kuersteiner. Semiparametric estimates of monetary policy effects: String theory revisited. *Journal of Business & Economic Statistics*, 36(3):371–387, 2018.
- S. Borağan Aruoba and Thomas Drechsel. Identifying monetary policy shocks: A natural language approach. *American Economic Journal: Macroeconomics*, forthcoming. NBER Working Paper 32417.
- Michael D. Bauer and Eric T. Swanson. An alternative explanation for the “Fed information effect”. *American Economic Review*, 113(3):664–700, 2023a.

- Michael D. Bauer and Eric T. Swanson. A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual*, 37:87–155, 2023b.
- Richard K. Crump, V. Joseph Hotz, Guido W. Imbens, and Oscar A. Mitnik. Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1):187–199, 2009.
- Alexander D’Amour, Peng Ding, Avi Feller, Lihua Lei, and Jasjeet Sekhon. Overlap in observational studies with high-dimensional covariates. *Journal of Econometrics*, 221(2):644–654, 2021.
- Deborah J. Danker and Matthew M. Luecke. Background on FOMC meeting minutes. *Federal Reserve Bulletin*, 91(2):175–179, 2005.
- Taeyoung Doh, Dongho Song, and Shu-Kuei Yang. Deciphering Federal Reserve communication via text analysis of alternative FOMC statements. *American Economic Journal: Macroeconomics*, 18(3):379–412, 2026.
- James D. Hamilton and Òscar Jordà. A model of the federal funds rate target. *Journal of Political Economy*, 110(5):1135–1167, 2002.
- Stephen Hansen, Michael McMahon, and Andrea Prat. Transparency and deliberation within the FOMC: A computational linguistics approach. *Quarterly Journal of Economics*, 133(2):801–870, 2018.
- Phillip Heiler and Ekaterina Kazak. Valid inference for treatment effect parameters under irregular identification and many extreme propensity scores. *Journal of Econometrics*, 222(2):1083–1108, 2021.
- Guido W. Imbens and Charles F. Manski. Confidence intervals for partially identified parameters. *Econometrica*, 72(6):1845–1857, 2004.
- Marek Jarociński and Peter Karadi. Deconstructing monetary policy surprises—the role of information shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43, 2020.
- Òscar Jordà. Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182, 2005.
- Shikun Ke, José Luis Montiel Olea, and James Nesbit. Robust machine learning algorithms for text analysis. Working paper, Cornell University, 2022.
- Shakeeb Khan and Elie Tamer. Irregular identification, support conditions, and inverse weight estimation. *Econometrica*, 78(6):2021–2042, 2010.
- Michal Kolesár and Mikkel Plagborg-Møller. Dynamic causal effects in a nonlinear world: The good, the bad, and the ugly. *Journal of Business & Economic Statistics*, 43(4):737–754, 2025.
- Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241, 1989.
- Toomas Laarits, Ben Matthies, Kaushik Vasudevan, and Will Yang. Policy by committee. Working paper 5364833, SSRN, 2025.
- Xinwei Ma and Jingshen Wang. Robust inference using inverse probability weighting. *Journal of the American Statistical Association*, 115(532):1851–1860, 2020.

- Charles F. Manski. Nonparametric bounds on treatment effects. *American Economic Review: Papers and Proceedings*, 80(2):319–323, 1990.
- Charles F. Manski. *Partial Identification of Probability Distributions*. Springer, New York, 2003.
- Matthew A. Masten and Alexandre Poirier. Identification of treatment effects under conditional partial independence. *Econometrica*, 86(1):317–351, 2018.
- Silvia Miranda-Agrippino and Giovanni Ricco. The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics*, 13(3):74–107, 2021.
- José Luis Montiel Olea and Mikkel Plagborg-Møller. Local projection inference is simpler and more robust than you think. *Econometrica*, 89(4):1789–1823, 2021.
- Emi Nakamura and Jón Steinsson. High-frequency identification of monetary non-neutrality: The information effect. *Quarterly Journal of Economics*, 133(3):1283–1330, 2018.
- Mikkel Plagborg-Møller and Christian K. Wolf. Local projections and VARs estimate the same impulse responses. *Econometrica*, 89(2):955–980, 2021.
- Ashesh Rambachan and Neil Shephard. When do common time series estimands have nonparametric causal meaning? Working paper, MIT and Harvard University, 2025. arXiv:1903.01637; revised version of “A Nonparametric Dynamic Causal Model for Macroeconometrics”.
- Christina D. Romer and David H. Romer. A new measure of monetary shocks: Derivation and implications. *American Economic Review*, 94(4):1055–1084, 2004.
- Christina D. Romer and David H. Romer. Presidential address: Does monetary policy matter? The narrative approach after 35 years. *American Economic Review*, 113(6):1395–1423, 2023.
- Christopher A. Sims. Macroeconomics and reality. *Econometrica*, 48(1):1–48, 1980.
- Christopher A. Sims. Interpreting the macroeconomic time series facts: The effects of monetary policy. *European Economic Review*, 36(5):975–1000, 1992.
- Herbert P. Susmann, Alec McClean, and Iván Díaz. Non-overlap average treatment effect bounds. Preprint 2509.20206, arXiv, 2025.

A Proofs

Proof of Proposition 1. By the law of iterated expectations, $\theta_d = \sum_x \pi_x(\mu_d(x) - \mu_0(x))$, and π_x is identified.

Identification of $\mu_a(x)$ for $a \in x$. By Assumption 2, $p_a(x, z) > 0$ for all z in the support of $Z \mid X = x$. By Assumption 1, $\mathbb{E}[\mathbf{1}\{D = a\}Y \mid X = x, Z] = \mathbb{E}[\mathbf{1}\{D = a\}Y(a) \mid X = x, Z] = p_a(x, Z) \mathbb{E}[Y(a) \mid X = x, Z]$. Dividing by $p_a(x, Z)$ and integrating over $Z \mid X = x$ gives (3).

Non-identification of $\mu_a(x)$ for $a \notin x$. By Assumption 2, $\mathbb{P}(D = a \mid X = x) = 0$, so the distribution of observables (Y, D, X, Z) places no restriction on the conditional distribution of $Y(a)$ given $X = x$ beyond Assumption 3. In particular, for any $m \in [K_1, K_2]$ the joint distribution obtained by setting $Y(a) = m$ almost surely on $\{X = x\}$ generates the same observables, satisfies

Assumption 1 (a constant is independent of D), and has $\mu_a(x) = m$. Hence the identified set for $\mu_a(x)$ is exactly $[K_1, K_2]$.

Sharpness of the sum. The unidentified means $\{\mu_a(x) : a \notin x\}$ enter θ_d linearly with coefficients $\pm\pi_x$, and each can be set independently to either endpoint by the construction above, applied menu by menu. The minimum of θ_d sets $\mu_d(x) = K_1$ for $x \in \mathcal{B}_d \cup \mathcal{N}_d$ and $\mu_0(x) = K_2$ for $x \in \mathcal{C}_d \cup \mathcal{N}_d$; the maximum does the reverse. This gives L_d and U_d . Every intermediate value is attained by convex combination, so the identified set is the interval $[L_d, U_d]$. Subtracting yields the width. $\square$

Proof of Proposition 2. Assumption 4 adds, for each $x \notin \mathcal{A}_d$, the constraint $\mu_d(x) - \mu_0(x) \in [\theta_d^A - \lambda, \theta_d^A + \lambda]$, where θ_d^A is identified. Intersecting with the cell-level intervals of Proposition 1 and summing gives the set. If no support constraint binds, each cell contributes $\pi_x(\theta_d^A \pm \lambda)$, so the bounds are $\pi(\mathcal{A}_d)\theta_d^A + (1 - \pi(\mathcal{A}_d))(\theta_d^A \pm \lambda) = \theta_d^A \pm \lambda(1 - \pi(\mathcal{A}_d))$, which excludes zero iff $\lambda < |\theta_d^A|/(1 - \pi(\mathcal{A}_d))$. $\square$

Information in X versus Z . If the menu contains all the information in Z that predicts D , i.e. $p_a(x, z) = p_a(x)$, then (3) becomes $\mu_a(x) = \mathbb{E}[Y \mid D = a, X = x]$, a difference in means within menu, and conditioning on Z is harmless. In general $\mathbb{E}[Y(a) \mid X = x] = \int \mathbb{E}[Y(a) \mid X = x, Z = z] dF_{Z|X=x}(z)$: the integral is over the distribution of Z *conditional on the menu*. The 2020 version integrated over the marginal distribution of Z , which is incorrect when X and Z are dependent, as they are here.

B Robustness: macro information set

Table 4 and Figure 4 repeat the main results with the macro information set of Angrist et al. (2018) (column (3) of Table 2) in place of the Greenbook forecasts.

Table 4: Effects of rate changes, macro information set

Policy	h	$\hat{\theta}_A$	90% CI	Identified set	λ^*
<i>Industrial production (%)</i>					
Rate increase	6	1.36	[0.78, 2.67]	[−2.9, 2.8]	4.11
	12	1.87	[0.62, 4.21]	[−4.5, 4.2]	5.67
	18	1.11	[−0.39, 3.54]	[−6.2, 4.3]	3.35
	24	0.38	[−1.41, 3.28]	[−7.1, 4.4]	1.14
Rate decrease	6	−0.41	[−1.40, 0.49]	[−5.7, 1.7]	1.02
	12	−1.68	[−3.85, −0.67]	[−9.1, 2.2]	4.14
	18	−2.33	[−5.60, −0.86]	[−11.0, 2.5]	5.73
	24	−2.52	[−6.62, −0.76]	[−11.7, 3.3]	6.21
<i>Unemployment rate (pp)</i>					
Rate increase	6	−0.35	[−0.65, −0.21]	[−0.6, 0.5]	1.05
	12	−0.73	[−1.41, −0.49]	[−1.1, 0.7]	2.21
	18	−0.52	[−1.01, −0.24]	[−1.1, 1.2]	1.56
	24	−0.57	[−1.30, −0.27]	[−1.1, 1.4]	1.74
Rate decrease	6	0.20	[−0.09, 0.42]	[−0.2, 1.2]	0.49
	12	0.73	[0.52, 1.34]	[−0.3, 2.1]	1.80
	18	1.13	[0.78, 2.19]	[−0.2, 2.8]	2.78
	24	1.27	[0.83, 2.56]	[−0.3, 3.0]	3.13
<i>PCE price index (%)</i>					
Rate increase	6	0.24	[0.08, 0.44]	[−0.9, 0.9]	0.71
	12	0.36	[0.08, 0.62]	[−0.9, 1.3]	1.09
	18	0.62	[0.21, 1.11]	[−0.7, 1.4]	1.86
	24	0.72	[0.20, 1.37]	[−0.9, 1.6]	2.17
Rate decrease	6	0.46	[0.23, 0.93]	[−1.0, 1.2]	1.13
	12	0.95	[0.52, 1.83]	[−0.8, 2.0]	2.34
	18	0.99	[0.49, 2.02]	[−0.7, 2.2]	2.44
	24	1.04	[0.45, 2.33]	[−0.9, 2.4]	2.55

Notes: As Table 3, with the propensity score of Table 2, column (3).

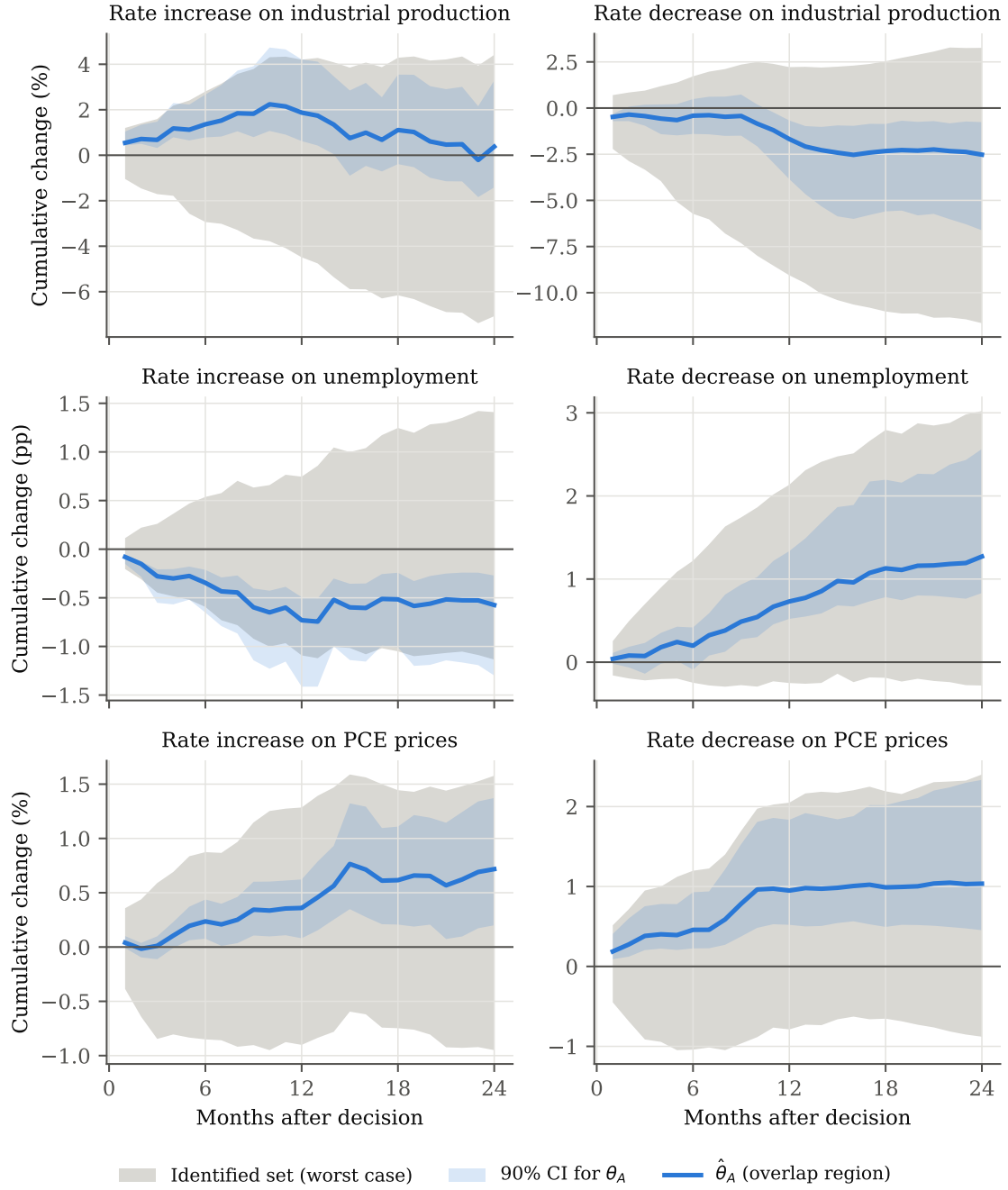


Figure 4: Cumulative effects of rate changes, macro information set. See Figure 2.

C Comparison with the 2020 version

The revision changes the estimation in several ways, described in the main text: the propensity score now imposes the menu’s zeros and uses covariates observed at the time of the meeting (Section 5); the bounds use the identified half of one-sided menus (Section 4); the outcome bounds K are calibrated to the range of the outcome rather than to the spread of the estimates; cumulative effects are bounded directly; and inference is reported. The 2020 estimation also applied a scaling step to the point-identified component of some bounds that is not part of the estimator derived in Section 4; the revision drops it. Most of the difference in the real-activity results between the two versions comes from this step.